

EQUIFLOW QA ASSESSMENT

BEFORE REPORT: Deliberate Red Run & Initial Test Case Failures

STATUS: 'L FAILED (Red Run Injected Bug & VIP Stale Assertions)

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1. Overview of Injected Regressions & Failures

As required by Stage 2 & Stage 3 of the Tactive Assessment, this document captures the BEFORE state of the test suite. Two failure conditions occurred: (1) A deliberate date-overlap regression injected into conflictEngine.ts to prove QA test effectiveness, and (2) Stale test assertions during Stage 3 VIP feature introduction before AI self-correction.

2. BEFORE Test Execution Matrix (With Failures)

Category	Test Scenario	Before Outcome
1. Happy Path	GET /api/assets -> Seeded inventory list	PASS [HTTP 200]
1. Happy Path	GET /api/health -> Health status check	PASS [HTTP 200]
1. Happy Path	VIP User auto-confirmed reservation	PASS [HTTP 201]
1. Happy Path	Standard User reservation -> PENDING queue	PASS [HTTP 201]
1. Happy Path	Admin can approve PENDING reservation	PASS [HTTP 200]
1. Happy Path	Admin can reject PENDING reservation	PASS [HTTP 200]
1. Happy Path	User can cancel own PENDING reservation	PASS [HTTP 200]
2. Business Rules	CONFLICT: Overlapping dates for same asset	FAIL [Expected 409, Got 201] 'L
2. Business Rules	QUOTA: Standard user blocked at >3 active bookings	PASS [HTTP 400]
2. Business Rules	BLACKOUT: Booking blocked during blackout window	PASS [HTTP 409]
2. Business Rules	PENALTY (VIP): 3 days overdue penalty fee	FAIL [Expected 90, Got 45] 'L
2. Business Rules	PENALTY (Standard): On-time return = \$0 penalty	PASS [HTTP 200]
3. Validation	REJECT: Inverted dates (start > end)	PASS [HTTP 409]
3. Validation	REJECT: Booking duration > 14 days limit	PASS [HTTP 409]
3. Validation	REJECT: Non-existent asset ID returns 404	PASS [HTTP 404]
3. Validation	REJECT: Unauthenticated request returns 401	PASS [HTTP 401]
3. Validation	REJECT: Standard user cannot create blackouts	PASS [HTTP 403]
3. Validation	REJECT: Missing required fields (Zod 400)	PASS [HTTP 400]
3. Validation	REJECT: Invalid date format (Zod 400)	PASS [HTTP 400]

3. Captured Terminal Output (Deliberate Red Run Failure)

